KENYA HIGH 2025 PREDICTION

KCSE TEST EXAM

233/3

CHEMISTRY

PAPER 3 (PRACTICAL)

TIME: 2¼ HOURS

NAME……………………………………………………………………………………

SCHOOL………………………………………………… **SIGN**………………………

INDEX NO……………………………………………… **ADM NO**……………………

(Kenya Certificate of Secondary Education)

CHEMISTRY PAPER 3 CONFIDENTIAL

INSTRUCTIONS

In additional to the apparatus and fittings common in a chemistry laboratory, each candidate will require the following.

- About 150ml of solution labelled **A**.
- About 100ml solution labelled **B**.
- About 50ml of solution labelled **C**
- About 0.2g of sodium hydrogen carbonate in a stoppered container.
- About 0.5g of solid **M** in a stoppered container.
- About 0.5g of solid **G** in a stoppered container
- 0 – 50ml burette.
- 25ml pipette.
- Two 250ml conical flasks
- 250ml volumetric flask
- 10ml measuring cylinder.
- Six test tubes on a test tube rack.
- A boiling tube.
- test tube holder.
- Complete stand.

- A white tile.
- One metallic spatula.
- Distilled water in a wash bottle.
- One label

Access to:

- Source of heat.
- Universal indicator paper and **its pH chart**.
- 2M aqueous ammonia supplied with a dropper.
- 2M aqueous sodium hydroxide supplied with a dropper.
- $Pb(NO_3)_{2\ (aq)}$ supplied with a dropper
- Acidified potassium manganate (VII) supplied with a dropper.
- Bromine water supplied with a dropper.
- 2M dilute nitric (V) acid.
- Methyl orange indicator with a dropper
- Phenolphthalein indicator with a dropper
- Sodium chloride solution

NB:

1. Solution **A** is prepared by dissolving 4.3 cm^3 of concentrated HCl (1.18g/cm^3) to 500 cm^3 of water and dilute to 1 litre.
2. Solution **B** is prepared by dissolving 1.2g of NaOH pellets in about 600ml of distilled water and diluting to 1 litre.
3. Solution **C** is prepared by dissolving 62.9g of $Na_2CO_3.10H_2O$ in about 800ml of distilled water and then topping up to 1 litre.
4. Acidified potassium permanganate is prepared by dissolving 3.5g of $KMnO_{4\ (s)}$ in 200cm^3 of 2M $H_2SO_{4(aq)}$ and toping up to one litre solution.
5. 2M $H_2SO_{4\ (aq)}$ is prepared by diluting 110cm^3 of concentrated Sulphuric (VI) acid to make one litre of solution.
6. 2M $NaOH_{(aq)}$ is prepared by dissolving 80g of NaOH pellets in one litre of solution.
7. 2M HNO_3 is prepared by adding 128 cm^3 of Conc. HNO_3 to about 500ml of water and dilute to 1 litre.
8. Sodium chloride solution is prepared dissolving 5.85g of NaCl in 1 litre of water
9. Lead (II) nitrate solution is prepared by dissolving 30g of $Pb(NO_3)_2$ in 1litre of water
10. Solid **M** is aluminium sulphate
11. Solid **G** is maleic acid.